

Multimodal YFB: Derivation and Updates

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1 Executive summary

Throughout, $i = 1, \dots, n$ indexes matched subjects, $m \in \{1, 2\}$ indexes modalities, $j = 1, \dots, p_m$ indexes features within modality m , and $k = 1, \dots, K$ indexes factors. The shorthand $\sum_{m,j}$ means $\sum_{m=1}^2 \sum_{j=1}^{p_m}$. When used in factor sums, a also ranges from 1 to K ; superscripts $-k$ and $-(m, j)$ indicate that factor k or feature (m, j) has been omitted.

This proposal defines a joint supervised EBMF–Cox model for matched expression and methylation. The modalities share subject scores L and survival coefficients β , while their loading matrices F_1, F_2 , point-exponential empirical-Bayes priors, and feature-wise residual precisions are modality-specific. The linear predictor $\eta_i = \sum_k \beta_k \sum_{m,j} y_{mij} f_{mjk}$ makes the Cox partial likelihood directly inform the loading updates. Thus, the model is not an unsupervised EBMF fit followed by a Cox regression: survival can influence which modality-specific loading patterns are retained.

With the Cox working quantities fixed, the main updates are:

- **Shared scores L .** Both reconstruction blocks contribute $A_{L_{ik}}$ and $B_{L_{ik}}$; their point-exponential EBNM update pools subjects within a factor. YFB has no direct survival term in this update.
- **Loadings F_m .** Each loading has reconstruction and survival contributions to $A_{F_{mjk}}$ and $B_{F_{mjk}}$, followed by a modality–factor-specific point-exponential EBNM update. Coordinates are updated sequentially because the survival predictor couples features.
- **Survival coefficients β .** Cox contributions define A_{β_k} and B_{β_k} ; the Normal prior gives a direct conjugate posterior update.
- **Residual precisions τ_{mj} .** Each feature receives the feature-wise maximum-likelihood update based on its expected squared reconstruction residual.

For all EBNM updates, the quadratic terms give the pseudo-observation $x = B/A$ with standard-error squared $s^2 = A^{-1}$. An intentionally over-specified K_{init} is reduced to an effective factor count K_{eff} through factor-wise empirical-Bayes shrinkage. The following sections state the full objective, the exact A/B expressions, the EBNM handoff, and the update order. The precision update τ is the one exception: it is a closed-form maximum-likelihood update, not an EBNM problem.

2 Model and priors

For matched subjects, let $Y_m \in \mathbb{R}^{n \times p_m}$ denote expression ($m = 1$) or methylation ($m = 2$). The extension stacks features while retaining one set of subject scores. Write $Y_m = (y_{mij})$ and $E_m = (E_{mij})$:

$$Y = [Y_1 \ Y_2], \quad F = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}, \quad Y_m = LF_m^\top + E_m. \quad (1)$$

- $L \in \mathbb{R}^{n \times K}$: shared subject scores.
- $F_m \in \mathbb{R}^{p_m \times K}$: modality-specific loadings.
- $E_{mij} \sim N(0, \tau_{mj}^{-1})$: independent Gaussian errors; each feature has its own precision.
- $\beta \in \mathbb{R}^K$: shared survival coefficients (the code's **B/EBeta**).

The **linear joint-YFB** survival predictor is

$$Z_{ik} = \sum_{m=1}^2 \sum_{j=1}^{p_m} y_{mij} f_{mjk}, \quad \boxed{\eta_i = \sum_k Z_{ik} \beta_k.} \quad (2)$$

Equivalently, the predictor for all subjects is

$$\boxed{\eta = (Y_1 F_1 + Y_2 F_2) \beta = Y F \beta,} \quad (3)$$

where $Y_m F_m$ is $n \times K$, so the sum is a shared subject-by-factor projection before multiplication by the K -vector β .

Unlike LB, whose predictor is $L\beta$, YFB depends directly on F and β . Thus survival contributes to their updates, but not directly to L .

Notation correspondence to the current single-modality YFB model. The posterior moments denoted in code by **EL**, **EL2**, **EF**, **EF2**, **EBeta**, **EBeta2**, and **Tau** correspond here to $\bar{L}$, $L^{(2)}$, $\bar{F}$, $F^{(2)}$, $\bar{\beta}$, $\beta^{(2)}$, and τ , respectively. The single-modality projection **ZF** = **Y %*% EF** is $\bar{Z}_{ik} = \sum_j y_{ij} \bar{f}_{jk}$; its multimodal analogue is $\bar{Z}_{ik} = \sum_{m=1}^2 \sum_j y_{mij} \bar{f}_{mjk}$. The code's partial residual **R_k** is the same object written here as R^{-k} , and its Cox working response $z_i = \eta_i^0 + u_i/w_i$ equals h_i/w_i when $w_i > 0$. Earlier single-mode derivations write the coefficient as $\tilde{\beta}$; it is the same quantity written here as β . The raw multimodal projection and its nonzero survival contribution to F are intentional proposed-model differences, not changes in the meanings of these shared symbols.

The empirical-Bayes priors are independent across entries, conditional on column-specific distributions:

$$\begin{aligned} l_{ik} &\sim g_{L_k} = (1 - \pi_{L_k}) \delta_0 + \pi_{L_k} \text{Exp}(r_{L_k}), \\ f_{mjk} &\sim g_{F_{mk}} = (1 - \pi_{F_{mk}}) \delta_0 + \pi_{F_{mk}} \text{Exp}(r_{F_{mk}}), \\ \beta_k &\sim g_{\beta_k} = N(0, s_{\beta_k}^2). \end{aligned} \quad (4)$$

Here π is the nonzero probability and r is the exponential rate. Every modality-factor pair has its own $(\pi_{F_{mk}}, r_{F_{mk}})$, fitted across that modality's features. The K score columns are shared, but a loading column may vanish in one modality. Hyperparameters are estimated; no hyperpriors or precision priors are introduced.

This proposal uses nonnegative point-exponential factors on a prespecified analysis scale. Each block must therefore be prepared on a documented nonnegative scale: for example, normalized nonnegative expression abundance and beta-scale methylation. Simulations must use the same

convention. A transformation that recenters features below zero is outside this formulation; using one would require signed factor priors and a new derivation. The raw linear projection in Equation 2 differs from the normalized single-modality YFB implementation and permits the joint loading update below.

The proposed default is the literal unweighted joint likelihood: each observation contributes once, with modality-specific precisions and priors. This does not force expression and methylation to have equal aggregate influence when $p_2 \gg p_1$. Feature-count imbalance will therefore be evaluated in simulation; inverse-feature-count weighting is a separate model extension, not an implicit default.

3 EBMF plus survival objective

Let t_i be observed survival time and δ_i the event indicator. At distinct event time t_r , write D_r for the event set, $d_r = |D_r|$, and R_r for the risk set; here r indexes distinct event times, whereas r_{L_k} and $r_{F_{mk}}$ above are exponential rates. The Breslow Cox partial likelihood is

$$\ell_C(\eta) = \sum_r \left\{ \sum_{i \in D_r} \eta_i - d_r \log \left(\sum_{i \in R_r} e^{\eta_i} \right) \right\}. \quad (5)$$

The EBMF observation likelihood becomes a sum over modalities:

$$\ell_Y = \sum_{m=1}^2 \left\{ \frac{n}{2} \sum_j \log \tau_{mj} - \frac{1}{2} \sum_{ij} \tau_{mj} \left(y_{mij} - \sum_k l_{ik} f_{mjk} \right)^2 - \frac{np_m}{2} \log(2\pi) \right\}. \quad (6)$$

The joint estimation criterion multiplies the observation likelihood, Cox partial likelihood, and independent priors:

$$\exp(\ell_Y + \ell_C) \left[\prod_{ik} g_{L_k}(l_{ik}) \right] \left[\prod_{mjk} g_{F_{mk}}(f_{mjk}) \right] \left[\prod_k g_{\beta_k}(\beta_k) \right].$$

The survival factor is a **partial likelihood**, not a full likelihood for survival times and a baseline hazard.

Use the mean-field approximation $q = \prod_{ik} q(l_{ik}) \prod_{mjk} q(f_{mjk}) \prod_k q(\beta_k)$. For fixed priors and precision, its objective is

$$\begin{aligned} \mathcal{J}(q, g, \tau) = & \mathbb{E}_q[\ell_Y] + \mathbb{E}_q[\ell_C] - \sum_{ik} \text{KL}(q(l_{ik}) \| g_{L_k}) \\ & - \sum_{mjk} \text{KL}(q(f_{mjk}) \| g_{F_{mk}}) - \sum_k \text{KL}(q(\beta_k) \| g_{\beta_k}). \end{aligned} \quad (7)$$

Relative to single-modality EBMF plus survival, the reconstruction term and loading-prior penalty now sum over modalities. There is still only one L prior term and one β prior term. All likelihood multipliers are one, so the F update below **adds** its reconstruction and survival terms rather than convex-combining them.

The current single-modality YFB implementation instead sets `alpha_F` = 0, suppressing the survival contribution to F because earlier dual-source updates were unstable. That stability safeguard is not part of the proposed multimodal model. Restoring stable survival feedback to the current single-modality YFB model is a separate model-development task.

Write $\bar{\theta} = \mathbb{E}_q \theta$, $\theta^{(2)} = \mathbb{E}_q \theta^2$, and $v_\theta = \theta^{(2)} - \bar{\theta}^2$. Each scalar update below isolates a quadratic $-A_\theta \theta^2/2 + B_\theta \theta$ and hence has the form

$$\boxed{x_\theta = B_\theta/A_\theta, \quad s_\theta^2 = A_\theta^{-1}, \quad q(\theta) \propto g(\theta)N(x_\theta; \theta, s_\theta^2).} \quad (8)$$

These are empirical-Bayes normal-means (EBNM) inputs. Second moments must be retained along with means. Updates with $A_\theta = 0$ are treated directly from the prior when $B_\theta = 0$, rather than dividing by zero. Here and below, C denotes terms constant with respect to the coordinate being updated.

4 Cox working approximation

The Cox partial likelihood in Equation 5 is the same Breslow partial likelihood used by the current single-modality YFB fitter, including its treatment of tied event times. At each event time, it compares the event subjects with the subjects still at risk; it does not require a parametric baseline hazard.

The Cox term is nonlinear in η , whereas the EBNM updates require a quadratic in the coordinate being updated. At each outer iteration, evaluate the current mean predictor η^0 and replace ℓ_C by a local weighted-least-squares surrogate. Hold that surrogate fixed during the next inner coordinate sweep. This is the same calculation that produces the single-mode working response z and weights w .

At event time t_r , P_{ri} is the normalized relative-risk weight for subject i within risk set R_r ; it is probability-like, but is not an additional model parameter. The score u_i is the observed event count less its current expected allocation, and w_i is a local-information weight. The full Cox Hessian couples subjects. The diagonal working approximation retains w_i while dropping those cross-subject terms, which permits scalar coordinate updates. Define these quantities by

$$P_{ri} = \frac{\mathbf{1}\{i \in R_r\}e^{\eta_i^0}}{\sum_{a \in R_r} e^{\eta_a^0}}, \quad u_i = \delta_i - \sum_r d_r P_{ri}, \quad w_i = \sum_r d_r P_{ri}. \quad (9)$$

The exact negative Hessian is $\sum_r d_r \{\text{diag}(P_r) - P_r P_r^\top\}$. We retain the existing **diagonal working approximation** $W = \text{diag}(w)$; w_i is not the exact Hessian diagonal. Replacing the curvature gives

$$\begin{aligned} Q_C(\eta \mid \eta^0) &= \ell_C(\eta^0) + u^\top(\eta - \eta^0) - \frac{1}{2}(\eta - \eta^0)^\top W(\eta - \eta^0) \\ &= C + \sum_i h_i \eta_i - \frac{1}{2} \sum_i w_i \eta_i^2, \quad h_i = u_i + w_i \eta_i^0. \end{aligned} \quad (10)$$

For $w_i > 0$, let $z_i^* = h_i/w_i = \eta_i^0 + u_i/w_i$; then $Q_C = C' - \frac{1}{2} \sum_i w_i (z_i^* - \eta_i)^2$. The division-free form using h_i also applies at zero curvature. Hold (w, h) fixed during a coordinate sweep. Earlier

single-mode derivations begin from this resulting (z, w) representation rather than deriving it; the present section makes that shared step explicit.

To take expectations, independence of loading entries gives

$$\bar{Z}_{ik} = \sum_{mj} y_{mij} \bar{f}_{mjk}, \quad v_{Z_{ik}} = \sum_{mj} y_{mij}^2 v_{f_{mjk}}, \quad Z_{ik}^{(2)} = \bar{Z}_{ik}^2 + v_{Z_{ik}}. \quad (11)$$

Independence of factors and of F from β then gives

$$\mu_{\eta_i} = \sum_k \bar{\beta}_k \bar{Z}_{ik}, \quad v_{\eta_i} = \sum_k \left\{ \beta_k^{(2)} v_{Z_{ik}} + v_{\beta_k} \bar{Z}_{ik}^2 \right\}. \quad (12)$$

Consequently,

$$\mathbb{E}_q Q_C = C + \sum_i h_i \mu_{\eta_i} - \frac{1}{2} \sum_i w_i (\mu_{\eta_i}^2 + v_{\eta_i}). \quad (13)$$

In the following updates, replace $\mathbb{E} \ell_C$ in Equation 7 by this expectation. The updates optimize the resulting **fixed working objective**; refreshing the expansion does not guarantee monotone improvement in the original nonquadratic variational objective.

4.1 From the working objective to an EBNM input

Let $\mathcal{J}_{\text{work}}$ denote Equation 7 after the replacement $\mathbb{E}_q[\ell_C] \leftarrow \mathbb{E}_q[Q_C]$. For a coordinate θ with prior g_θ , the coordinate-ascent rule is

$$\log q^*(\theta) = \mathbb{E}_{-\theta} [\ell_Y + Q_C] + \log g_\theta(\theta) + C. \quad (14)$$

Each section below first removes the terms of this expression that do not contain the coordinate. Algebra then collects the remaining likelihood terms as

$$\begin{aligned} \log q^*(\theta) &= -\frac{A_\theta}{2} \theta^2 + B_\theta \theta + \log g_\theta(\theta) + C \\ &= \log [g_\theta(\theta) N(x_\theta; \theta, s_\theta^2)] + C', \\ x_\theta &= B_\theta / A_\theta, \quad s_\theta = A_\theta^{-1/2}. \end{aligned} \quad (15)$$

Thus the likelihood algebra ends once A_θ and B_θ have been identified: (x_θ, s_θ) are the normal-means data supplied to EBNM. For a collection of coordinates sharing one prior, an implementation uses the corresponding vectors in `ebnm(x = x, s = s, prior_family = ...)` and extracts the posterior mean and standard deviation; the required second moment is $\mathbb{E}_q(\theta^2) = \text{sd}_q(\theta)^2 + \mathbb{E}_q(\theta)^2$.

For L_k , this is one vector EBNM call across subjects. For F_k within a modality, the Cox term couples feature coordinates, so a single simultaneous vector call is not the exact coordinate algorithm. Instead, update the pseudo-observations sequentially using a shared current $g_{F_{mk}}$, then re-estimate that shared prior from the completed feature sweep. For β_k , the stated Normal prior makes the EBNM posterior available in closed form; an EBNM implementation can produce the same normal-means posterior. The τ_{mj} update has no prior and therefore does not enter EBNM.

5 Update the shared scores L

Start from the fixed working objective. Replace $\mathbb{E}_q[\ell_C]$ in Equation 7 by $\mathbb{E}_q[Q_C]$ from Equation 13. For coordinate l_{ik} , this Cox contribution is constant because $\eta = YF\beta$ contains F and β , but not L . Thus only the reconstruction term and the L prior contribution remain. Define the reconstruction residual excluding factor k ,

$$R_{mij}^{-k} = y_{mij} - \sum_{a \neq k} l_{ia} f_{mja}, \quad \bar{R}_{mij}^{-k} = y_{mij} - \sum_{a \neq k} \bar{l}_{ia} \bar{f}_{mja}. \quad (16)$$

The retained objective terms are

$$[\mathcal{J}_{\text{work}}]_{l_{ik}} = \mathbb{E}_q \left[-\frac{1}{2} \sum_{mj} \tau_{mj} (R_{mij}^{-k} - l_{ik} f_{mjk})^2 \right] - \text{KL}\{q(l_{ik}) \| g_{L_k}\} + C. \quad (17)$$

Applying the coordinate rule Equation 14 to this expression gives

$$\log q^*(l_{ik}) = -\frac{1}{2} \sum_{mj} \tau_{mj} \mathbb{E}_{-l_{ik}} [(R_{mij}^{-k} - l_{ik} f_{mjk})^2] + \log g_{L_k}(l_{ik}) + C. \quad (18)$$

Here $q^*(l_{ik})$ is the coordinate-optimal variational factor and $\mathbb{E}_{-l_{ik}}$ takes expectation over every variational variable except l_{ik} .

Expand and take expectations. Terms depending on l_{ik} are

$$-\frac{1}{2} \sum_{mj} \tau_{mj} \left\{ l_{ik}^2 f_{mjk}^{(2)} - 2l_{ik} \bar{f}_{mjk} \bar{R}_{mij}^{-k} \right\}.$$

The cross-product factors because R^{-k} excludes factor k . Both modalities therefore contribute information about the same score.

Collect the quadratic and hand off to EBNM. Substituting the preceding expansion into Equation 18 gives

$$\log q^*(l_{ik}) = -\frac{A_{L_{ik}}}{2} l_{ik}^2 + B_{L_{ik}} l_{ik} + \log g_{L_k}(l_{ik}) + C,$$

where the coefficients and the resulting normal-means input are

$$\boxed{\begin{aligned} A_{L_{ik}} &= \sum_{m=1}^2 \sum_j \tau_{mj} f_{mjk}^{(2)}, \\ B_{L_{ik}} &= \sum_{m=1}^2 \sum_j \tau_{mj} \bar{f}_{mjk} \bar{R}_{mij}^{-k}, \\ x_{L_{ik}} &= B_{L_{ik}} / A_{L_{ik}}, \quad s_{L_{ik}}^2 = A_{L_{ik}}^{-1}. \end{aligned}} \quad (19)$$

For a fixed column k , the subject updates separate. Estimate its prior using the EBNM marginal likelihood

$$\hat{g}_{L_k} = \arg \max_{g \in \mathcal{G}_{\text{PE}}} \sum_i \log \int N(x_{L_{ik}}; l, s_{L_{ik}}^2) dg(l), \quad (20)$$

then obtain $\bar{l}_{ik}$ and $l_{ik}^{(2)}$ from Equation 8 with $\hat{g}_{L_k}$. Here $\mathcal{G}_{\text{PE}}$ is the point-exponential family in Equation 4.

What changes: the sums include all modality blocks. The EBNM step and absence of a direct survival term in the YFB score update are unchanged. There is no separate score matrix for methylation.

6 Update modality-specific loadings F_m

Fix a modality $m \in \{1, 2\}$ ($m = 1$ for expression and $m = 2$ for methylation), a feature $j \in \{1, \dots, p_m\}$ within that modality, and a factor $k \in \{1, \dots, K\}$. Thus f_{mjk} is a loading in modality m and uses the modality-factor prior $g_{F_{mk}}$.

Start from the fixed working objective. Let ℓ_{Y_m} denote the m th summand of ℓ_Y in Equation 6. In the fixed working objective, the terms involving $f = f_{mjk}$ are precisely the reconstruction term for modality m , the working Cox term, and its prior contribution:

$$[\mathcal{J}_{\text{work}}]_{f_{mjk}} = \mathbb{E}_q \left[-\frac{1}{2} \tau_{mj} \sum_i (R_{mij}^{-k} - l_{ik} f_{mjk})^2 + Q_C \right] - \text{KL}\{q(f_{mjk}) \| g_{F_{mk}}\} + C. \quad (21)$$

Applying Equation 14, equivalently retaining only the terms that contain f_{mjk} , gives

$$\log q^*(f_{mjk}) = [\mathbb{E}_q \ell_{Y_m}]_{f_{mjk}} + [\mathbb{E}_q Q_C]_{f_{mjk}} + \log g_{F_{mk}}(f_{mjk}) + C. \quad (22)$$

Reconstruction contribution. Isolating $f = f_{mjk}$ yields $-\frac{1}{2} \tau_{mj} \sum_i \mathbb{E}(R_{mij}^{-k} - l_{ik} f)^2$. Expanding gives $-A_Y f^2/2 + B_Y f$, where

$$A_Y = \tau_{mj} \sum_i l_{ik}^{(2)}, \quad B_Y = \tau_{mj} \sum_i \bar{l}_{ik} \bar{R}_{mij}^{-k}. \quad (23)$$

Working-survival contribution. Separate other factors from other features of this factor:

$$\begin{aligned} \eta_i &= \eta_i^{-k} + \beta_k(y_{mij}f + Z_{ik}^{-(m,j)}), & \eta_i^{-k} &= \sum_{a \neq k} \beta_a Z_{ia}, \\ Z_{ik}^{-(m,j)} &= Z_{ik} - y_{mij}f_{mjk}, & \bar{Z}_{ik}^{-(m,j)} &= \bar{Z}_{ik} - y_{mij}\bar{f}_{mjk}. \end{aligned} \quad (24)$$

Let $h_i^{-k} = h_i - w_i \bar{\eta}_i^{-k}$, where $\bar{\eta}_i^{-k} = \sum_{a \neq k} \bar{\beta}_a \bar{Z}_{ia}$. Thus, $\bar{Z}_{ik}^{-(m,j)}$ is the mean projection for factor k with feature j of modality m omitted. In Equation 10, the terms depending on f are

$$\sum_i h_i^{-k} \bar{\beta}_k y_{mij} f - \frac{1}{2} \sum_i w_i \beta_k^{(2)} \{y_{mij}^2 f^2 + 2y_{mij} f \bar{Z}_{ik}^{-(m,j)}\}. \quad (25)$$

The second cross-product uses $\mathbb{E}\beta_k^2$, since both features multiply the **same** random coefficient. It includes features from both modalities.

Combine the two likelihood contributions. The reconstruction and working survival expressions give

$$\begin{aligned} \log q^*(f) &= -\frac{1}{2}(A_Y + A_C)f^2 + (B_Y + B_C)f + \log g_{F_{mk}}(f) + C, \\ A_C &= \beta_k^{(2)} \sum_i w_i y_{mij}^2, \\ B_C &= \sum_i y_{mij} \{\bar{\beta}_k h_i^{-k} - w_i \beta_k^{(2)} \bar{Z}_{ik}^{-(m,j)}\}. \end{aligned} \quad (26)$$

Identify the EBNM input and apply the modality's prior. Thus $A_{F_{mjk}} = A_Y + A_C$ and $B_{F_{mjk}} = B_Y + B_C$, giving

$$\begin{aligned} A_{F_{mjk}} &= \tau_{mj} \sum_i l_{ik}^{(2)} + \beta_k^{(2)} \sum_i w_i y_{mij}^2, \\ B_{F_{mjk}} &= \tau_{mj} \sum_i \bar{l}_{ik} \bar{R}_{mij}^{-k} \\ &\quad + \sum_i y_{mij} \{\bar{\beta}_k h_i^{-k} - w_i \beta_k^{(2)} \bar{Z}_{ik}^{-(m,j)}\}, \\ x_{F_{mjk}} &= B_{F_{mjk}} / A_{F_{mjk}}, \quad s_{F_{mjk}}^2 = A_{F_{mjk}}^{-1}. \end{aligned} \quad (27)$$

Use Equation 8 with $g_{F_{mk}}$ to obtain the two posterior moments. Survival couples loading coordinates through $\bar{Z}^{-(m,j)}$. Therefore update coordinates sequentially using current moments, holding the prior fixed during the sweep; independent simultaneous featurewise EBNM fits would not be this coordinate algorithm.

Thus, unlike an unsupervised EBMF fit followed by Cox regression, the Cox partial likelihood directly informs the loading patterns selected in both modality blocks.

6.1 Estimate each loading prior

After updating the loadings of modality m and factor k , hold their posteriors fixed and maximize the expected log prior. Let $\omega_{mjk} = q(f_{mjk} > 0)$ be the posterior slab probability. For $\pi = \pi_{F_{mk}}$ and $r = r_{F_{mk}}$, the relevant terms are

$$\sum_j \{\omega_{mjk} \log \pi + (1 - \omega_{mjk}) \log(1 - \pi)\} + \left(\sum_j \omega_{mjk} \right) \log r - r \sum_j \bar{f}_{mjk}. \quad (28)$$

Differentiating with respect to π and r gives

$$\frac{\sum_j \omega_{mjk}}{\pi} - \frac{p_m - \sum_j \omega_{mjk}}{1 - \pi} = 0, \quad \frac{\sum_j \omega_{mjk}}{r} - \sum_j \bar{f}_{mjk} = 0.$$

Thus the empirical-Bayes coordinate maximization is

$$\boxed{\pi_{F_{mk}} \leftarrow \frac{1}{p_m} \sum_j \omega_{mjk}, \quad r_{F_{mk}} \leftarrow \frac{\sum_j \omega_{mjk}}{\sum_j \bar{f}_{mjk}}.} \quad (29)$$

These equations are applied **separately to expression and methylation**. If the posterior slab mass is zero, the fitted distribution is δ_0 ; the unused slab rate is unidentified. The next loading sweep uses the updated prior. A posterior/prior sweep may be repeated to convergence at fixed working curvature.

Why this prior step differs from the score step. Given other factors, the L likelihood separates over subjects, allowing a vector EBNM fit. The survival likelihood for F couples features. Alternating scalar posterior updates and the prior maximization above respects that coupling. Each scalar posterior uses a common modality-factor prior, not a separately estimated prior for every loading.

What changes from a single modality: each loading gets information from its own reconstruction block and the shared survival projection; prior estimation pools only features in its own modality. Setting the survival contribution to zero would produce an omics-only loading update, which is not the joint model derived here.

7 Update the shared survival coefficients β

Start from the fixed working objective. In that objective, only the working Cox term and the β_k prior contribution depend on β_k . Write $\eta_i = \eta_i^{-k} + Z_{ik}\beta_k$. The working survival term and g_{β_k} are the only terms involving β_k :

$$[\mathcal{J}_{\text{work}}]_{\beta_k} = \mathbb{E}_q[Q_C] - \text{KL}\{q(\beta_k) \| g_{\beta_k}\} + C, \quad (30)$$

where the displayed expectation retains only its β_k -dependent terms. Applying Equation 14 gives

$$\log q^*(\beta_k) = \sum_i \left\{ h_i^{-k} \mathbb{E} Z_{ik} \beta_k - \frac{1}{2} w_i \mathbb{E} Z_{ik}^2 \beta_k^2 \right\} + \log g_{\beta_k}(\beta_k) + C. \quad (31)$$

Here Z_{ik} is independent of η_i^{-k} under the mean-field approximation. The second moment follows from Equation 11, including loading uncertainty in both modalities.

Identify coefficients.

$$\begin{aligned}
A_{\beta_k} &= \sum_i w_i \left(\bar{Z}_{ik}^2 + v_{Z_{ik}} \right), \\
B_{\beta_k} &= \sum_i \bar{Z}_{ik} h_i^{-k}, \\
x_{\beta_k} &= B_{\beta_k} / A_{\beta_k}, \quad s_{\text{obs},k}^2 = A_{\beta_k}^{-1}.
\end{aligned} \tag{32}$$

Equivalently, the coordinate expression is

$$\log q^*(\beta_k) = -\frac{A_{\beta_k}}{2} \beta_k^2 + B_{\beta_k} \beta_k + \log g_{\beta_k}(\beta_k) + C,$$

which is the scalar normal-means input in Equation 15. The next step uses the chosen Normal prior to write its posterior analytically.

Estimate the prior and obtain posterior moments. With prior variance $s_{\beta_k}^2$, the scalar marginal distribution is $x_{\beta_k} \sim N(0, s_{\text{obs},k}^2 + s_{\beta_k}^2)$. Differentiating its log likelihood with respect to the total variance gives $-1/(2v) + x_{\beta_k}^2/(2v^2) = 0$. Enforcing a nonnegative prior variance gives

$$\hat{s}_{\beta_k}^2 = \max\{0, x_{\beta_k}^2 - s_{\text{obs},k}^2\}. \tag{33}$$

For $\hat{s}_{\beta_k}^2 > 0$, completing the square in $-A_{\beta_k} \beta_k^2/2 + B_{\beta_k} \beta_k - \beta_k^2/(2\hat{s}_{\beta_k}^2)$ gives

$$v_{\beta_k} = \left(A_{\beta_k} + \hat{s}_{\beta_k}^{-2} \right)^{-1}, \quad \bar{\beta}_k = v_{\beta_k} B_{\beta_k}, \quad \beta_k^{(2)} = v_{\beta_k} + \bar{\beta}_k^2. \tag{34}$$

At zero fitted prior variance, $q(\beta_k) = \delta_0$. The coefficient remains shared; there is no expression-specific or methylation-specific β_k . Replacing $Z^{(2)}$ by $\bar{Z}^2$ would omit loading uncertainty and would not yield this variational update.

8 Update feature-wise precisions τ_m

Start from the fixed working objective. The precision τ_{mj} appears only in the observation likelihood for feature j of modality m ; there is no precision prior in this proposal. For a single feature, define $S_{mij} = \mathbb{E}_q[(y_{mij} - \sum_k l_{ik} f_{mjk})^2]$. The precision-dependent objective is

$$\mathcal{J}(\tau_{mj}) = \frac{n}{2} \log \tau_{mj} - \frac{\tau_{mj}}{2} \sum_i S_{mij} + C. \tag{35}$$

Expand and take expectations. For $X_{mij} = \sum_k l_{ik} f_{mjk}$, $\mathbb{E}(y - X)^2 = (y - \mathbb{E}X)^2 + \text{Var } X$. Independence between L and F , and across factor columns, implies

$$\begin{aligned}
\mathbb{E}X_{mij} &= \sum_k \bar{l}_{ik} \bar{f}_{mjk}, \\
\text{Var } X_{mij} &= \sum_k \left\{ l_{ik}^{(2)} f_{mjk}^{(2)} - \bar{l}_{ik}^2 \bar{f}_{mjk}^2 \right\}.
\end{aligned}$$

Therefore the expected squared residual is

$$S_{mij} = \left(y_{mij} - \sum_k \bar{l}_{ik} \bar{f}_{mjk} \right)^2 + \sum_k \left\{ l_{ik}^{(2)} f_{mjk}^{(2)} - \bar{l}_{ik}^2 \bar{f}_{mjk}^2 \right\}. \quad (36)$$

The variance term can equivalently be written $\sum_k \{v_{l_{ik}} \bar{f}_{mjk}^2 + v_{f_{mjk}} \bar{l}_{ik}^2 + v_{l_{ik}} v_{f_{mjk}}\}$. This form makes its nonnegativity explicit.

Differentiate and solve.

$$\frac{\partial \mathcal{J}}{\partial \tau_{mj}} = \frac{n}{2\tau_{mj}} - \frac{1}{2} \sum_i S_{mij} = 0, \quad \frac{\partial^2 \mathcal{J}}{\partial \tau_{mj}^2} = -\frac{n}{2\tau_{mj}^2} < 0.$$

For positive expected residual sum of squares,

$$\tau_{mj} \leftarrow \frac{n}{\sum_i S_{mij}}, \quad m = 1, 2, \quad j = 1, \dots, p_m. \quad (37)$$

This is a point maximum-likelihood update, not a Gamma posterior update. It has the same form as single-modality EBMF, applied independently to features in each block. If the expected residual sum is zero, no finite MLE exists; an implementation must report that boundary rather than silently presenting a capped value as the MLE.

9 Complete algorithm

Inputs: matched training matrices (Y_1, Y_2) , survival (t, δ) , chosen K , and numerical tolerances. Feature order and preprocessing are recorded for subsequent prediction.

1. **Initialize.** Choose finite score/loading moments, positive precisions, and priors with positive slab mass and rate; initialize β away from an absorbing zero state, then compute Equation 11.
2. **Refresh the survival approximation.** Set $\eta^0 = \mu_\eta$, compute (u, w, h) from Equation 9 and Equation 10, and freeze them for the inner updates.
3. **For each factor k , update in order:**
 - Form residuals excluding k using current moments (Equation 16).
 - Update $q(L_{\cdot k})$ and g_{L_k} using Equation 19 and Equation 20.
 - For each modality m , update $q(f_{mjk})$ sequentially using Equation 27, refreshing affected projection moments after each coordinate; then update $g_{F_{mk}}$ using Equation 29.
 - Recompute the current other-factor predictor and update g_{β_k} and $q(\beta_k)$ using Equation 32 through Equation 34.
4. **Update noise.** Compute expected residuals and update all τ_{mj} using Equation 36 and Equation 37.
5. **Assess convergence.** At fixed (w, h) , sweep until the working objective and moments stabilize, then refresh the approximation. Stop when outer iterations stabilize the predictor and moments; record the fixed objective and mean-predictor Cox partial likelihood, not $\mathbb{E}_q \ell_C$.

Do not normalize columns during fitting: factor scales are exchangeable. After convergence, reporting may rescale a nonzero score column with the corresponding loadings and coefficient.